

HARSH GUPTA

GENERATIVE AI ENGINEER

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SUMMARY

Generative AI-focused Software Engineer with 2+ years of experience in full-stack Python development, building AI-powered solutions. Skilled in LangChain, LLM integration, prompt engineering, React.js, Node.js, and modern databases, with strong problem-solving and debugging abilities.

EXPERIENCE

Generative AI Engineer

07/2024 - Present

Tata Consultancy Services

- Designed and implemented LLM-powered automation workflows that reduced manual data processing efforts by 40% for internal stakeholders.
- Built LangChain-based AI pipelines to integrate Generative AI capabilities into enterprise web applications and automate business workflows.
- Developed scalable full-stack applications using React.js and Node.js, improving frontend bundle size by 25% and significantly enhancing application performance.
- Designed and developed high-performance RESTful APIs capable of supporting 500+ concurrent requests while maintaining 99.9% service availability.

ReactJS Developer

01/2023 - 07/2023

SA Auditors & Consultants LLC

- Developed responsive and scalable web applications using React.js, modern JavaScript (ES6+), and reusable component.
- Built modular UI components that improved maintainability and reduced code duplication across projects.
- Implemented centralized state management using Redux for predictable application state.
- Developed a real-time chat application using React.js, Node.js, Express.js, and Socket.IO for low-latency communication.
- Integrated Firebase Authentication to enable secure user authentication and session management.

SKILLS & AWARDS

Gen AI LangChain | LLM | Prompt | Engineering | LangGraph | Pinecone | RAG | NLP | Agentic | AI | MCP | Chroma

Database Python | JavaScript | TypeScript | Java | C++ | HTML5 | CSS3 | React.js | Next.js

PROJECTS

AI-Powered Shorts Generator

Python | LangChain | Gemini AI | MoviePy | Kyutai TTS | Pollinations AI

- Built an end-to-end Generative AI content generation pipeline that automated YouTube Shorts production, reducing video creation time from hours to minutes.
- Designed multi-stage AI workflows using LangChain for script generation, prompt orchestration, and conversational content creation.
- Leveraged Gemini AI and advanced prompt engineering techniques to generate engaging multi-character conversations with dynamic personas.
- Integrated Kyutai TTS voice cloning to produce natural multi-speaker voiceovers and automated audio stitching.
- Engineered a custom video generation engine using MoviePy, synchronized subtitles, gameplay footage, animated visuals, and AI-generated audio.
- Built a scalable workflow combining multiple AI services into an automated content generation pipeline.

Chat App

React.js | Node.js | Express.js | Socket.IO | Firebase

- Built a real-time messaging platform using Socket.IO with sub-50ms message delivery.
- Implemented secure user authentication using Firebase Authentication.
- Developed scalable WebSocket infrastructure supporting instant messaging and online presence detection.

Stock Dekho

Python | Django | Pandas | Plotly | JavaScript

- Developed a stock market analytics platform to visualize historical market data.
- Built interactive candlestick charts using Plotly for market trend analysis.
- Processed stock market datasets using Pandas and Yahoo Finance API.

EDUCATION

Chandigarh University BCA CGPA: 8.05	2021 - 2024
Bhartiya Vidya Mandir Sen Sec School, Ludhaina, Punjab, India CBSE (Class XII) Percentage: 86.6%	2020 - 2021
Shifaly Public School School, Ludhiana, Punjab, India PSEB (Class X) Percentage: 81.5%	2018 - 2019